The Curvature-Fidelity Discriminator

Why reach does not separate (Doc. 305)

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Doc. 305 The curvature-fidelity discriminator — why reach does not separate

Abstract. Doc. 304 narrows the distinction between a uniform and an accumulating memory to a choice of axis: what separates is not *reach* (finite window versus unbounded integral) but *curvature-fidelity* (does the accumulator carry the $1/n$-decaying pitch, the self-similar texture, the ratio e). This note supports the finding internally to FFGFT with a deterministic testbench: at *equal reach*, a reach-based metric returns the same for both accumulators, while the e-criterion (Doc. 295) separates them cleanly and exactly — self-similarity $D(2k) - D(k) \to \ln 2$, pitch decay $d_n \sim n^{-1}$, scale ratio $\to e$. The finding is derivable from ξ / the recursion (*proven*). It establishes *no* specificity of any cognitive memory model; that extension remains a restricted bridge (Category B) and requires a separate, independent, jointly sealed benchmark.

.1 The reduced point

In Doc. 304 the operational question of a "genuine" memory collapses onto a single axis. A memory that carries something about a past sits, in FFGFT, at $D(k)$, the cumulated defect of the unrolled time-winding

$$D(k) = \sum_{n \le k} 100\, \xi_n \;\to\; \ln\!\left(1 + \frac{k}{75}\right), \qquad \xi_{n+1} = \xi_n\,(1 - 100\,\xi_n) \tag{.1}$$

(Doc. 295/296). Two accumulators can then be contrasted: one that carries the fractal *curvature* of this winding (the $1/n$-decaying pitch), and one that *linearises* it (constant pitch, Archimedean winding, the $d = 0$ limit). Doc. 304 argues that the load-bearing distinction is not reach but which of the two forms the accumulator carries. This note tests exactly that.

.2 Two accumulators of equal reach

The testbench (`ffgft_kruemmungstreue_probe.py`) contrasts both forms at *equal reach*, i.e. with the same total accumulation $D(N)$ at the horizon N:

- *curvature-faithful* — the exact recursion $d_{n+1} = d_n(1 - d_n)$ with $d_n = 100\,\xi_n$, start $d_1 = 1/75$; cumulated $D(k) \to \ln(1 + k/75)$ (log-spiral, ratio e);
- *uniform* — constant pitch $d_{\text{const}} = D(N)/N$, cumulated $D_{\text{lin}}(k) = d_{\text{const}}\,k$ (Archimedean, no self-similarity).

Both reach the same D at N — reach is identical by construction.

.3 Reach does not separate

An *unbounded* accumulator keeps the contribution of an arbitrarily old event in its running sum; a finite kernel of width W loses it once $N - t_0 > W$. A reach-based test — "does an early impulse still affect the output?" — therefore separates {accumulators} from {finite kernel}, but returns the same for the curvature-faithful and the uniform accumulator: both retain the impulse. Reach *cannot* tell the two apart.

.4 Curvature-fidelity separates (the e-criterion)

The e-criterion of Doc. 295 — a winding carries ratio e exactly when the defect per step decays as $1/n$ — separates them cleanly. Three signatures, all checked against the exact recursion:

- *Self-similarity.* The scale-doubling increment $D(2k) - D(k)$ is, for the curvature-faithful accumulator, *scale-invariant*, $\to \ln 2 \approx 0.6931$ (at $k = 10^5$: 0.69274, deviation 4.1×10^{-4}); for the uniform one it grows linearly with k (at $k = 10^5$: 2.76). Only the log-spiral is similar to itself across scales.
- *Pitch decay.* The exponent p in $d_n \sim n^{-p}$ is, curvature-faithful, ≈ 1 (log-log slope -0.984 over a decade, i.e. $1/n$); uniform, $p = 0$ (constant).
- *Scale ratio.* The k-factor per unit increase of D converges, curvature-faithful, to e (at $D = 6$: 2.7226, deviation 4.3×10^{-3}; from $\exp D = 1 + k/75$); uniform, to 1 (arithmetic, no log-spiral).

The convergence is asymptotically exact (an $O(1/k)$ remainder, consistent with the congruence/self-similarity discussion in Doc. 304): the self-similarity is not a finite exact symmetry but a continuous one that settles fully in the limit.

.5 The positive criterion

Hence the positive discriminator Doc. 304 calls for: what is to be tested is not "finite window versus accumulation" (reach) but *curvature-fidelity at equal reach* — does the accumulator carry the $1/n$ pitch, the ratio e, the self-similar texture? A long but linear kernel passes every reach test and remains uniform; a short but curvature-faithful accumulator already carries the structure. The e-criterion is thus sharper than any reach boundary and makes the axis operationally testable.

.6 Status

Proven (internal to FFGFT). The accumulation form, the three-signature separation, and the e-criterion are derivable from ξ and the recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ and verified against the exact recursion (Doc. 295; script `ffgft_kruemmungstreue_probe.py`, deterministic). The $O(1/k)$ asymptotics is named.

Restricted (Category B). This testbench establishes *no* specificity of a cognitive, operationally and functionally defined memory (recursive access, semantic compression, affective weighting, identity binding, persistent state carry, predictive reuse). Whether curvature-fidelity separates such an object remains the restricted bridge of Doc. 304 and requires a separate, *independent, jointly sealed* benchmark — not an FFGFT-internal calculation. In both directions: a system may carry the $1/n$ texture without those functions, and a functionally relevant memory is not ruled out merely for failing the e-criterion.

Provenance and symmetry. $D(k)$, the e-criterion, and the FFGFT interpretation of temporal curvature originate in the FFGFT corpus; the operational memory architecture against which a bridge would be tested originates in independent work (Doc. 304, provenance section). Neither framework contains, grounds, or subordinates the other; rejection of the bridge leaves both unchanged.